Shanshan Luo （罗姗姗）

Contact Information

Department of Applied Statistics
School of Mathematics and Statistics
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Employment

January 2026 - present *Associate Professor*
School of Mathematics and Statistics, Beijing Technology and Business University, Beijing, China

September 2022 - December 2025 *Lecturer*
School of Mathematics and Statistics, Beijing Technology and Business University, Beijing, China

Education

September 2017 - July 2022 *Ph.D. in Statistics*
School of Mathematical Sciences, Peking University, Beijing, China.
Advisor: Prof. Yangbo He

September 2013 - July 2017 *B.S. in Mathematics*
School of Mathematical Sciences, Capital Normal University, Beijing, China.

January 2026 - June 2026 *Visiting Scholar*
Department of Pure Mathematics and Mathematical Statistics, University of Cambridge, Cambridge, UK.
Host: Prof. Qingyuan Zhao

Research Interests

My research primarily focuses on causal inference, with specific interest in the following areas:

1. Causal effect: covariate adjustment, data fusion, instrumental variables, measurement error, principal stratification, propensity scores, spillover effects
2. Causal attribution: individual attribution analysis, continuous outcome attribution
3. Causal discovery: Bayesian networks, causal mechanisms of latent confounders, proximal variable selection
4. Missing data: nonignorable missing data

Publications

1. Yafang Deng[†], Kang Shuai, and Shanshan Luo[*]. Bidirectional causal inference for binary outcomes in the presence of unmeasured confounding. To appear in *Statistica Sinica*, 2026.
2. Shanshan Luo, Wei Li, Xueli Wang, Shaojie Wei[*], and Zhi Geng. Assessing interactive causes of an occurred outcome due to two binary exposures. To appear in *Biometrics*, 2026.
3. Jiaqi Min[†], Xueyue Zhang[†], and Shanshan Luo[*]. A regression-based approach for bidirectional proximal causal inference. To appear in *Journal of the Royal Statistical Society: Series A*, 2026.
4. Shanshan Luo, Mengchen Shi[†], Wei Li[*], Xueli Wang, and Zhi Geng. Efficiency-improved doubly robust estimation with non-confounding predictive covariates. *Electronic Journal of Statistics*, 2025; 19(2): 3723–3742.
5. Shanshan Luo and Zhi Geng[*]. Discussion on "Causal and Counterfactual Views of Missing Data Models". To appear in *Statistica Sinica*, 2025.

6. Wei Li, Yuan Liu, Shanshan Luo*, and Zhi Geng. Causal inference with outcomes truncated by death and missing not at random. To appear in *Statistics in Medicine*, 2025.
7. Shanshan Luo, Yixuan Yu†, Chunchen Liu, Feng Xie*, and Zhi Geng. Causal attribution analysis for continuous outcomes. *ICML*, Vancouver, Canada, 2025. (Spotlight, top 2.6%)
8. Shanshan Luo#, Yechi Zhang#, Wei Li*, and Zhi Geng. Multiply robust estimation of causal effects using linked data. *Computational Statistics & Data Analysis*, 2025; 209: 108175.
9. Peng Wu, Shanshan Luo*, and Zhi Geng. On the comparative analysis of average treatment effects estimation via data combination. *Journal of the American Statistical Association*, 2025; 120(552): 2250–2261.
10. Shanshan Luo, Jiaqi Min†, Wei Li, Xueli Wang*, and Zhi Geng. A comparative analysis of different adjustment sets using propensity score based estimators. *Computational Statistics & Data Analysis*, 2025; 203: 108079.
11. Shaojie Wei, Chao Zhang, Zhi Geng, and Shanshan Luo*. Identifiability and estimation for potential-outcome means with misclassified outcomes. *Mathematics*, 2024; 12(18): 2801.
12. Shanshan Luo, Wei Li*, Wang Miao, and Yangbo He*. Identification and estimation of causal effects in the presence of confounded principal strata. *Statistics in Medicine*, 2024; 43(22): 4372–4387.
13. Kang Shuai, Shanshan Luo*, Wei Li, and Yangbo He. Identifying causal effects using instrumental variables from the auxiliary population. To appear in *Statistica Sinica*, 2024.
14. Kang Shuai, Shanshan Luo, Yue Zhang, Feng Xie, and Yangbo He*. Identification and estimation of causal effects using non-Gaussianity and auxiliary covariates. To appear in *Statistica Sinica*, 2024.
15. Feng Xie, Zhengming Chen, Shanshan Luo*, Wang Miao, Ruichu Cai, and Zhi Geng. Automating the selection of proxy variables of unmeasured confounders. *ICML*, Vienna, Austria, 2024. (Spotlight, top 3.5%)
16. Honglei Zhang, Shuyi Wang, Haoxuan Li, Chunyuan Zheng, Xu Chen, Li Liu, Shanshan Luo*, and Peng Wu*. Uncovering the limitations of eliminating selection bias for recommendation: missing mechanisms, disentanglement, and identifiability. *ICDE*, Utrecht, Netherlands, 2024.
17. Wei Li, Shanshan Luo, and Wangli Xu*. Calibrated regression estimation using empirical likelihood under data fusion. *Computational Statistics & Data Analysis*, 2024; 190: 107871.
18. Wei Li, Shanshan Luo*, Yangbo He, and Zhi Geng. Subgroup analysis using Bernoulligated hierarchical mixtures of experts models. *Statistics in Medicine*, 2023; 42(26): 4681–4695.
19. Shanshan Luo, Wei Li*, and Yangbo He. Causal inference with outcomes truncated by death in multiarm studies. *Biometrics*, 2023; 79(1): 502–513.

*Corresponding author, #Co-first author, †Supervised student.

Working Papers

1. Shanshan Luo, Yafang Deng†, Qingyuan Zhao, and Zhi Geng. Apportioning causal responsibility of two risk factors for an adverse outcome via counterfactual attribution. *arXiv*, *Revision*, 2026.
2. Shanshan Luo, Yingying Wang†, Mengchen Shi†, and Yue Zhang. Improved inverse probability weighting estimation and its application to predictive covariates. *Revision*, 2026.

3. Naiwen Ying, Ping Zhang, Shanshan Luo, and Wang Miao. A generalized tetrad constraint for testing conditional independence given a latent variable. *arXiv*, *Revision*, 2026.
4. Shanshan Luo, Kang Shuai, Yechi Zhang, Wei Li, and Yangbo He. Identification and estimation of causal peer effects using instrumental variables. *arXiv*, *Revision*, 2026.
5. Yue Zhang, Shanshan Luo, Zhi Geng, and Yangbo He. Optimal treatment rules under missing predictive covariates: a covariate-balancing doubly robust approach. *arXiv*, 2025.
6. Peng Wu, Qing Jiang, and Shanshan Luo. Safe individualized treatment rules with controllable harm rates. *arXiv*, 2025.
7. Shanshan Luo, Peng Wu, and Zhi Geng. Policy learning with pseudo-type classification. *arXiv*, 2025.
8. Yingying Wang[†], Yuan Liu, and Shanshan Luo. Selecting among missingness models for sequential outcomes with nonignorable nonresponse. *arXiv*, 2026.
9. Yue Zhang, Shanshan Luo, and Yangbo He. Orthogonal policy learning with ordinal outcomes. *arXiv*, 2026.
10. Yi Qin[†], and Shanshan Luo. Partial identification and inference for the probability of necessity using instrumental variables (in Chinese), submitted, 2026.

Awards Grants

Outstanding Graduate of Beijing, 2017.

National Scholarship, 2021.

Outstanding Graduate of Beijing, 2022.

Young Scientists Fund (Type C) of the National Natural Science Foundation of China (NSFC), January 2025 - December 2027.

Young Elite Scientists Sponsorship Program of the Beijing High Innovation Plan, Beijing Association for Science and Technology, December 2025 - December 2027.

Teaching Experience

Applied Stochastic Processes: Fall 2022

Multivariate Statistical Analysis: Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025

Causal Inference: Spring 2023, Fall 2023, Fall 2024, Spring 2025

Probability Theory and Mathematical Statistics: Fall 2026

Student Supervision

2023: Jiaqi Min (Master's, graduated July 2026); Xueyue Zhang (Master's, graduated July 2025).

2024: Yafang Deng (Master's, graduated July 2026); Zhiqiang Li (Master's, graduated July 2026); Taiyan Liu (Master's, graduated July 2026); Yingying Wang (Master's, expected July 2027).

2025: Jia Lin (Master's, expected 2028); Wenshuo Xu (Master's, expected 2028); Zhiyuan Zhu (Master's, expected 2028).

2026: Yi Qin (Ph.D., expected 2030); Tianyi Bai (Master's, expected 2029); Mengfan Xu (Master's, expected 2029); Zhen Zhang (Master's, expected 2029).